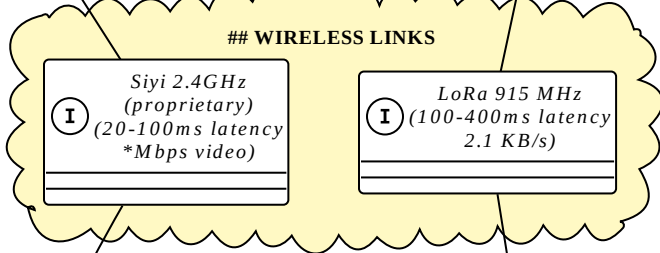
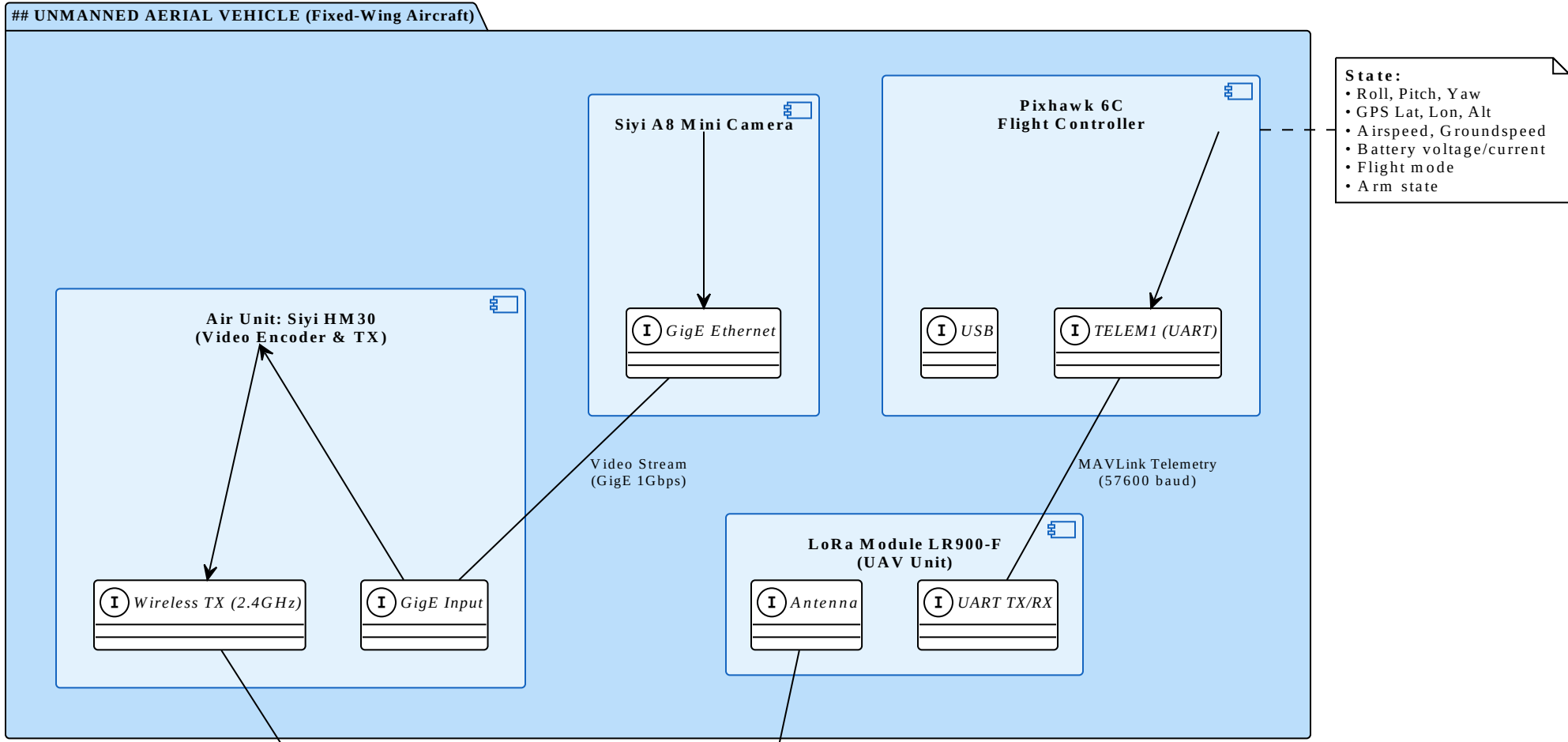


Autonomous Fixed-Wing UAV GCS Architecture
Pixhawk 6C + LoRa + Siyi Video + ROS2 + Mission Planner



Telemetry Path (DOWN - UAV → GCS):

- Pixhawk 6C → UART → LoRa TX → (915MHz) → LoRa RX → COM3/USB → MAVProxy → TCP:14550 (Mission Planner) TCP:14551 (ROS2)

Command Path (UP - GCS → UAV):

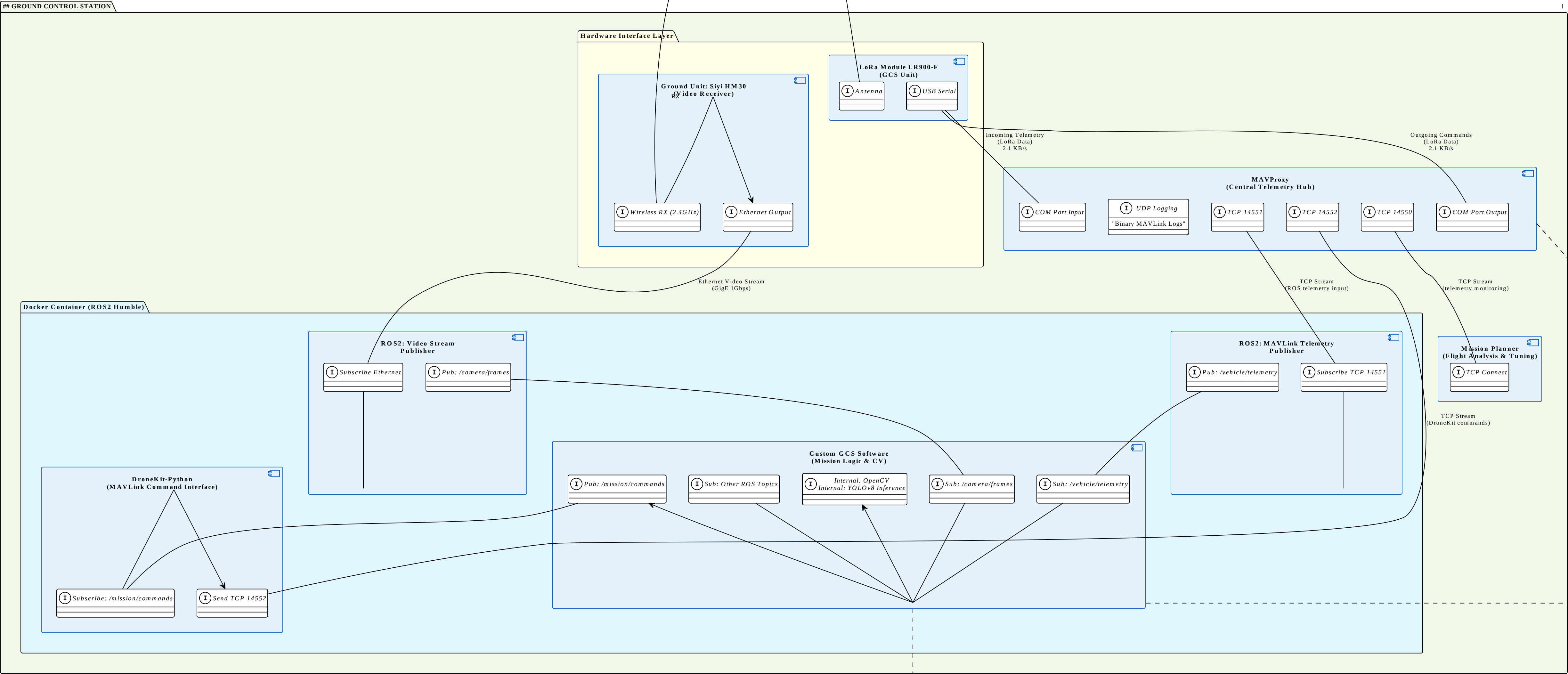
- Custom GCS → DroneKit → TCP:14552 → MAVProxy → COM3/USB → LoRa TX → (915MHz) → LoRa RX → UART → Pixhawk 6C

Video Path (DOWN - UAV → GCS):

- Siyi A8 mini → GigE → Siyi Air HM30 → (2.4GHz proprietary) → Siyi Ground HM30 → Ethernet → ROS2 Video Publisher

Latency Budget:

- LoRa telemetry: 100-400ms round-trip
- Video: 20-100ms (Siyi proprietary)
- ROS2 processing: 50-200ms
- Total mission loop: 200-700ms



Critical Note: Multiplexes telemetry from LoRa to multiple clients (Mission Planner, ROS2, DroneKit) without serial port conflicts

Inputs:

- Pixhawk telemetry
- Video frames (GigE)
- Internal state

Processing:

- OpenCV frame processing
- YOLOv8 inference
- Decision logic

Outputs:

- Arm/Disarm
- Set Flight mode
- Send waypoints
- Payload commands

Autonomous Mission Brain
Subscribes to telemetry + video
Runs CV inference pipeline
Publishes control commands via DroneKit